



Khulna University of Engineering & Technology

Department of Computer Science & Engineering

Project Report on

Hall Dining Management System

Course title: Information Systems Design Laboratory
Course code: CSE 3120
Submission Date: October 19, 2025

Submitted to,

Md Nazirulhasan Shawon
Lecturer,
Department of Computer Science and
Engineering

Safin Ahmmed
Lecturer,
Department of Computer Science and
Engineering

Submitted by,

Name : Saleheen Uddin Sakin
Roll : 2107103

Name : Abdullah Al Noman
Roll : 2107105

Name : Sree Shuvo Kumar Joy
Roll : 2107116

Group : B2
Year : 3rd
Semester : 1st
Department: Computer Science and
Engineering

Objectives:

- To learn about Software Development Life Cycle.
- To learn about the design patterns and how to implement them in system development.
- To learn about Git and GitHub and maintain git while building a system.
- To follow Agile Method for project management using Jira tool.
- To understand various types of graphs and diagrams and use them in project development.

Introduction:

The Problem: Tracking daily dining hall expenses across paper receipts and multiple spreadsheets is inefficient and makes it hard to maintain accurate financial control.

Project Goal: To create a simple web app that helps a dining manager easily track daily expenses and manage their dining operations more effectively.

Our Solution: A clean, centralized website where the manager can log in, record every purchase, and see their spending in one place.

Key Features:

The system was designed with specific feature sets for each user role:

- **For Managers:**
 - **Secure Login/Logout:** Role-based access to a dedicated dashboard.
 - **Expense Tracking:** Log and categorize daily shopping costs and other expenses.
 - **Meal Menu Management:** Set and publish the daily meal menu (Regular, Manager Special, Feast).
 - **View Summary:** Can see an overview of spending and meal statistics.
- **For Students:**
 - **View Meal Menu:** Check the published menu for each day.
 - **Meal On/Off Toggle:** A simple interface to indicate their participation for upcoming meals, allowing for accurate headcounts.
 - **View Summary:** Can see an overview of spending and meal statistics.
- **For Administrators:**
 - **User Management:** Create new user accounts and assign roles (Student, Manager, Admin).

- **Generate Reports:** Create summary reports of total spending and meal attendance over a defined period.

Methodology:

The system's architecture was planned using standard design models to ensure a clear and structured development process.

System Design Models:

Use Case Diagram: Illustrated interactions between actors (Manager, Student, Admin) and all system features.

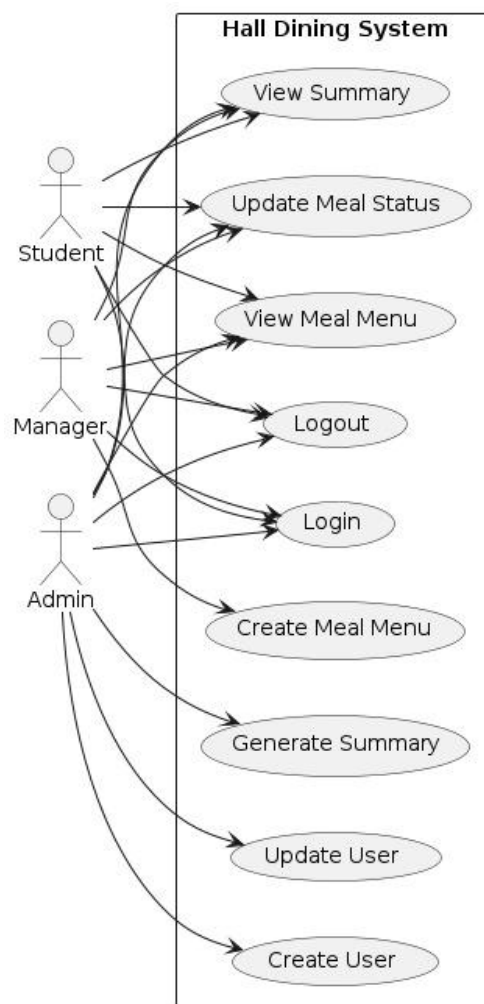


Fig: Use Case Diagram of the hall management System

Entity-Relationship (ER) Diagram: Served as the direct blueprint for the MongoDB database schema.

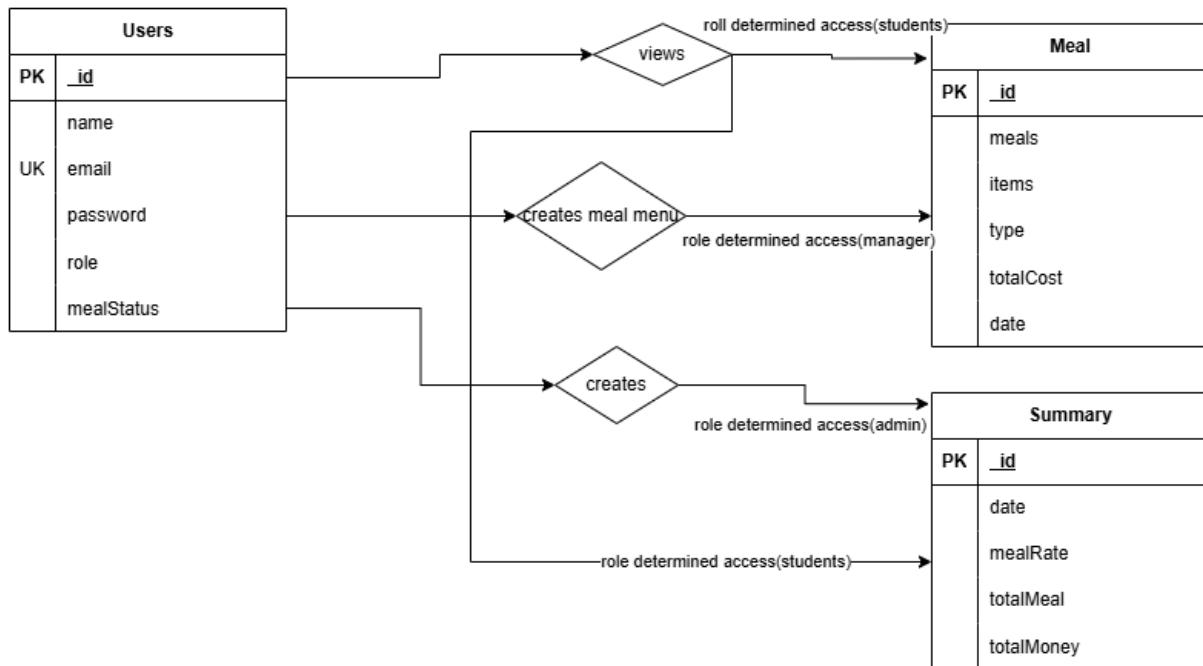


Fig: ER diagram for the hall management System

Activity Diagram: Modeled the business workflow for operations like "Recording an Expense."

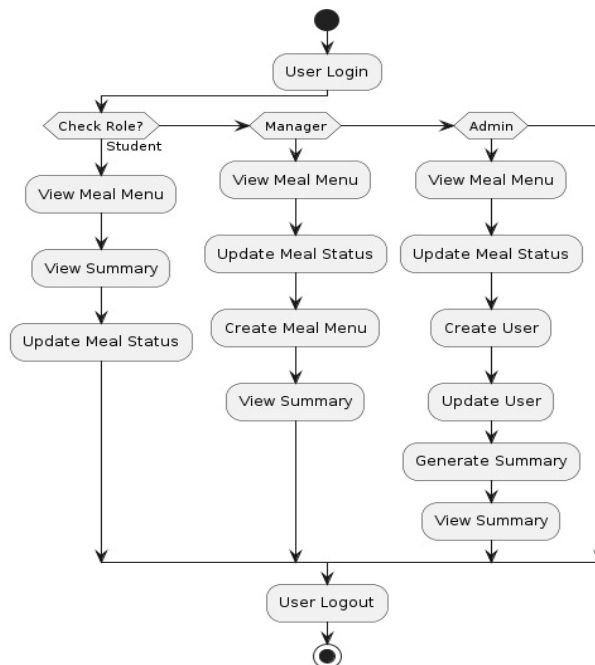


Fig: Activity Diagram for hall dining management.

Sequence Diagram: Detailed the step-by-step object interactions for key processes like user authentication.

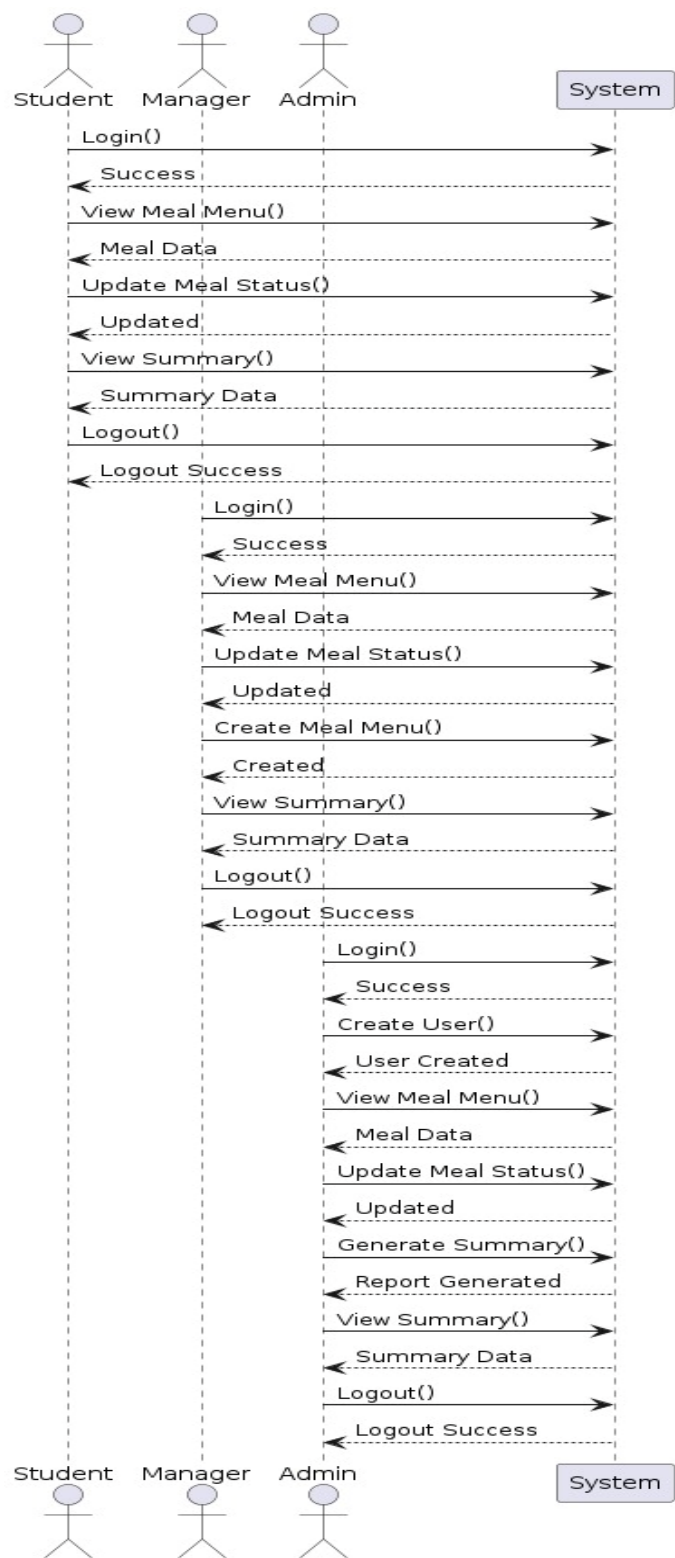


Fig: Sequence Diagram for the hall management system

Gantt Chart: Provided a high-level visual timeline for project phases and deadlines.

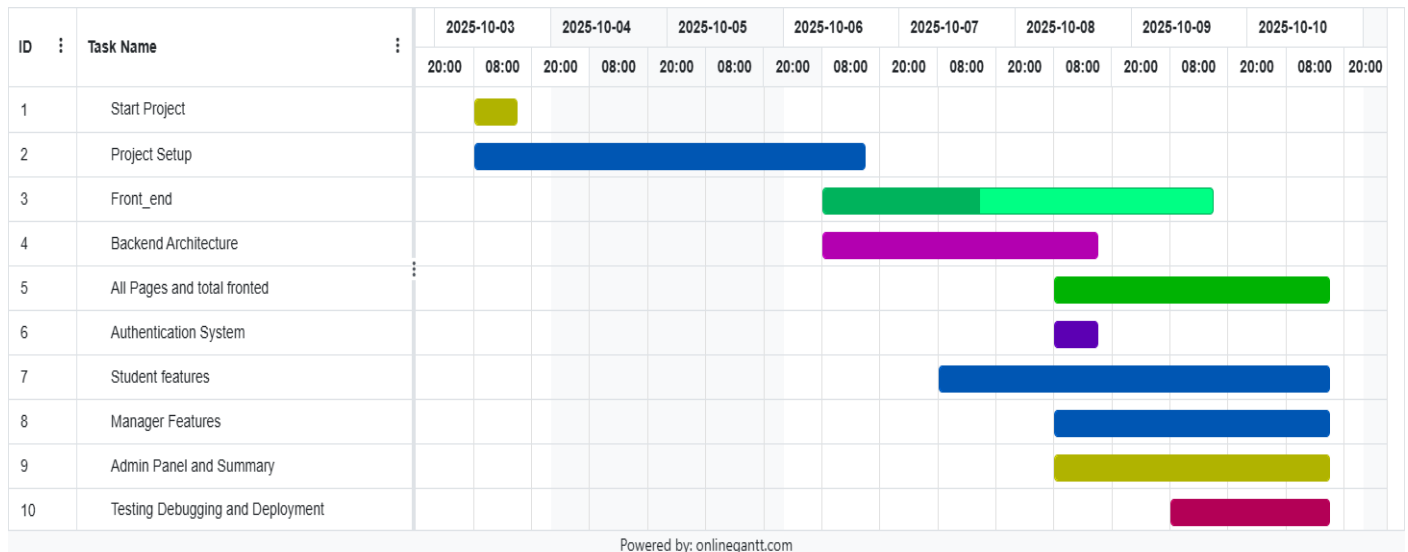


Fig: Gantt Chart for better tracking.

Data Flow Diagram: For the better comprehension of the data flow in the system

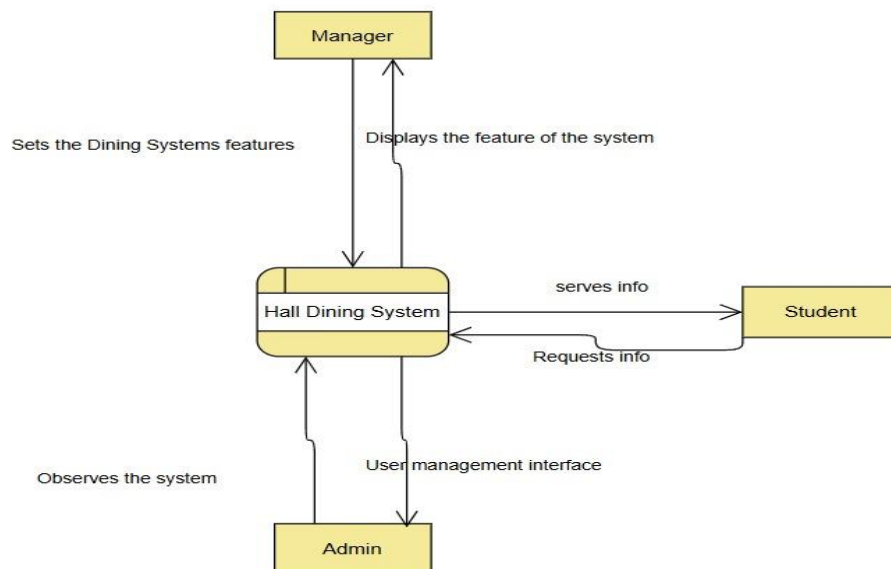


Fig: DFD(context level) for Hall Management System

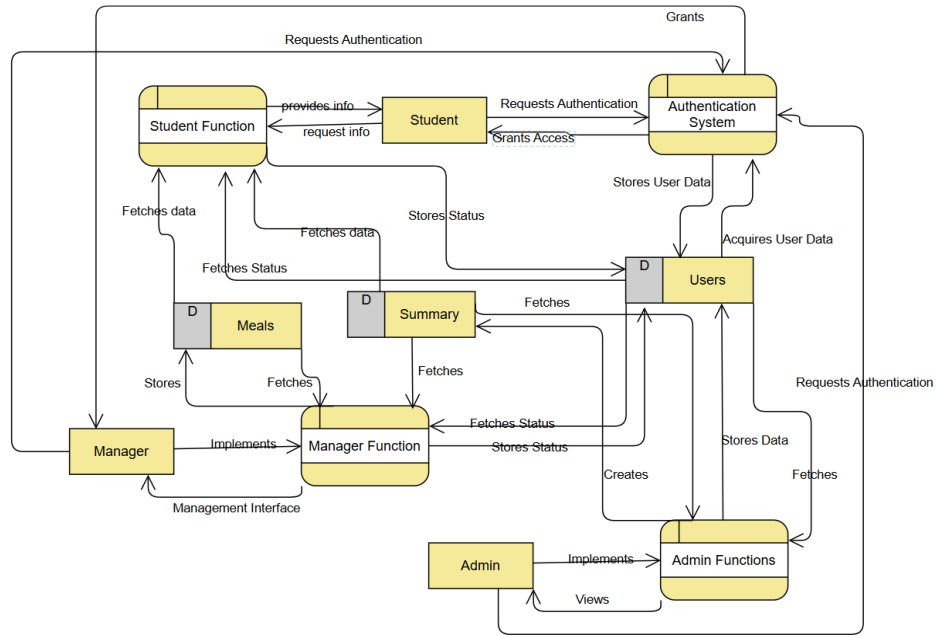


Fig : Level 1 DFD

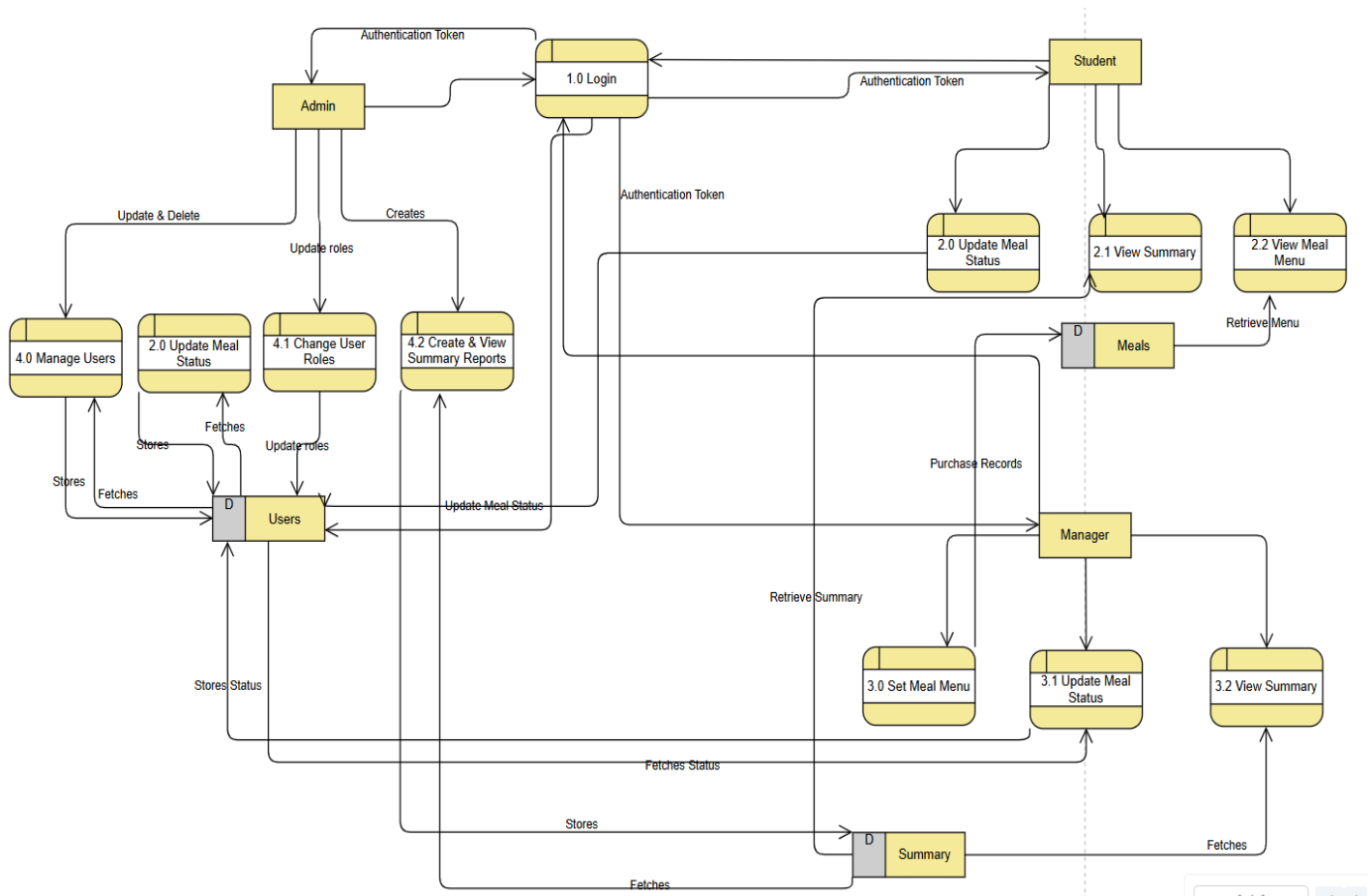


Fig: Level 2 DFD

Implementation & Development:

Technology Stack:

- **Frontend:** React.js
- **Backend:** Express.js
- **Language:** TypeScript
- **Database:** MongoDB
- **Project Management:** Jira
- **Version Control:** GitHub
- **Deployment:** Vercel & Netlify

SDLC & Sprint Overview:

- **Sprint 1: Project Foundation & Backend Setup**
 - Initialized GitHub repository and Jira project board.
 - Set up Express.js server with TypeScript and established connection to MongoDB.
 - Developed core database models (User, Meal) and implemented a basic login system with backend authentication.
- **Sprint 2: Core Feature Development & Integration**
 - Expanded backend APIs to include operations for Meals, Expenses, and User management.
 - Developed key frontend components and integrated them with the backend APIs.
 - Implemented role-based login and routing, directing users to appropriate dashboards.
- **Sprint 3: UI Finalization, Deployment & Testing**
 - Completed the remaining frontend user interfaces with a focus on responsive UI/UX design.
 - Conducted comprehensive integration testing and user acceptance testing (UAT).
 - Deployed the final application to a live production environment.

Testing:

A multi-faceted testing strategy was employed to ensure application reliability:

- **Unit Testing:** Verified individual backend functions and frontend components.
- **Integration Testing:** Ensured seamless communication between frontend and backend APIs.

Team Roles & Contributions:

- **2107103: Backend & Deployment Lead**
 - Architected and developed the entire backend system, including RESTful APIs and database models.
 - Implemented authentication and security middleware.
 - Managed the final deployment process and led the testing efforts.
- **2107116: Scrum Master & Frontend Developer**
 - Facilitated Agile ceremonies, maintained the Jira backlog, and ensured team productivity.
 - Contributed significantly to frontend development, building several key components.
- **2107105: UI/UX Designer & Frontend Lead**
 - Designed the overall wireframes, mockups, and user experience flow.
 - Created the website pages using React and modular technique

Discussion & Challenges:

The Agile methodology proved highly effective, allowing the team to adapt to challenges such as refining API specifications between sprints. The initial division of backend and frontend tasks in parallel sprints ensured steady progress. A key challenge was managing state for the meal toggle feature and ensuring real-time data consistency, which was resolved through careful API design and frontend state management. The use of Jira and GitHub created a transparent and collaborative environment.

Conclusion

The Hall Dining Manager System was successfully delivered as a robust, secure, and user-friendly web application that meets all core objectives. It effectively centralizes dining hall operations, providing managers with financial control, students with convenience, and administrators with oversight. The project stands as a testament to the effective application of modern web technologies and agile principles, providing a solid foundation for future enhancements such as push notifications, advanced analytics, and mobile application development.